

	Network Administration <i>Build an extended virtual network: Testing Static and Dynamic ARP</i>	2025/2026
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Objectives :

- Build an extended virtual network and verify connectivity together with ARP

This lab has been extended from the original PC1/PC2-only setup to include additional PCs. All PCs are now connected via Linux bridges (br-lan1 and br-lan2), which simulate Ethernet switches, enabling full LAN connectivity within each subnet. This lab demonstrates LAN connectivity using Linux network namespaces and virtual bridges, and focuses on observing and comparing dynamic vs static ARP behavior.

Network scenario and requirements

Background / Scenario

This lab simulates two LANs using Linux bridges:

- LAN1: 192.168.10.0/24 → PC1, PC3, PC5
- LAN2: 192.168.20.0/24 → PC2, PC4, PC6
- Linux bridges (br-lan1 and br-lan2) act as Ethernet switches connecting all PCs.
- A GATEWAY namespace connects both LANs, while the ISP (209.165.200.224/27) is simulated separately.
- PCs route through the GATEWAY; LANs are isolated from the ISP unless routing/NAT is enabled (Lab3).

Lab Topology (Virtual)

Device	Interface	IP Address	Network/Mask
LAN 1 PCs	veth-PC1-pc	192.168.10.10	192.168.10.0/24
	veth-PC3-pc	192.168.10.11	192.168.10.0/24
	veth-PC5-pc	192.168.10.12	192.168.10.0/24
LAN 2 PCs	veth-PC2-pc	192.168.20.10	192.168.20.0/24
	veth-PC4-pc	192.168.20.11	192.168.20.0/24
	veth-PC6-pc	192.168.20.12	192.168.20.0/24
GATEWAY	LAN1 (gw-lan1)	192.168.10.1	192.168.10.0/24
	LAN2 (gw-lan2)	192.168.20.1	192.168.20.0/24
	WAN (veth3)	209.165.200.226	209.165.200.224/27
ISP	WAN (isp1)	209.165.200.225	209.165.200.224/27

Part 1: Build a virtual network

Prepare a Bash script that automatically creates, configures, and cleans up the abovementioned network topology. The script, as in Lab0, must include:

- Clean-up mechanism
- Namespace creation
- Virtual link creation
- IP addressing
- Interface and loopback activation
- Routing configuration
- Gateway behavior

Part 2: Connectivity testing

Once you've set up your virtual network, there are several Linux commands you can use to inspect and verify the configuration.

Step 1: Verify namespaces exist

```
# Verify that all namespaces were created correctly.
```

```
ip netns list
```

```
# You should see:
```

```
PC1-PC6 SW1 SW2 GATEWAY ISP
```

Step 2: Verify interfaces are UP

```
# Avoid false failures caused by down interfaces.
```

```
sudo ip netns exec PC1 ip link
```

```
sudo ip netns exec SW1 ip link
```

Step 3: Verify IP addressing

```
# Confirm correct Layer-3 configuration.Repeat for other PCs.
```

```
sudo ip netns exec PC1 ip addr show veth-PC1-pc
```

Step 4 : Test local connectivity (same LAN)

```
# Verify Layer 2 + ARP + ICMP. Do for Lan2
```

```
sudo ip netns exec PC1 ping -c 3 192.168.10.12
```

Step 5: Test connectivity to the gateway

```
# Verify switches correctly separate broadcast domains.  
sudo ip netns exec PC1 ping -c 3 192.168.20.10
```

Because this lab first focuses on ARP and Layer-2 behavior, disabling IP forwarding prevents routing between LANs, making ARP resolution, broadcast domains, and LAN isolation clearly observable. IP forwarding is re-enabled later to study routing and NAT.

Step 6: Disable IP forwarding on GATEWAY

```
# Add this after GATEWAY is created  
sudo ip netns exec GATEWAY sysctl -w net.ipv4.ip_forward=0
```

Step 7: Inter-LAN routing blocked

```
# FAILS  
sudo ip netns exec PC1 ping 192.168.20.10
```

The output shows that the ping fails, because we need pure Layer-2 switching and ARP, with no Layer-3 routing anywhere.

Part 3: ARP & switching observations

Step 1: Start with a clean ARP state

```
# On PC1  
sudo ip netns exec PC1 ip neigh flush all  
# Output should be empty  
sudo ip netns exec PC1 ip neigh
```

Step 2: Observe Dynamic ARP creation

```
# Trigger ARP by pinging another host in the same LAN.  
sudo ip netns exec PC1 ping -c 2 192.168.10.11  
# Check the output  
sudo ip netns exec PC1 ip neigh
```

Step 3: Observe ARP aging (Dynamic behavior)

Wait ~30–60 seconds without traffic. Dynamic ARP ages out automatically.

```
sudo ip netns exec PC1 ip neigh
```

Step 4: View ARP traffic on the wire

On PC1

```
sudo ip netns exec PC1 tcpdump -i veth-PC1-pc arp
```

In another terminal

```
sudo ip netns exec PC1 ping 192.168.10.12
```

Check your ARP table on PC1

```
sudo ip netns exec PC1 ip neigh
```

You will see two-way dynamic ARP learning in action

- PC1 asks: “Who has 192.168.10.12?” → this is a broadcast ARP request.
- PC5 (192.168.10.12) replies with its MAC address.
- PC5 now asks PC1 for its MAC.
- PC1 replies with its MAC.

Step 5 : Create a Static ARP entry

Manually add a static ARP entry on PC1 for PC3

```
sudo ip netns exec PC1 ip neigh add 192.168.10.11 lladdr a:bd:60:a8:29:e7 dev veth-PC1-pc nud permanent
```

Find PC3 MAC first

```
sudo ip netns exec PC3 ip link show veth-PC3-pc
```

If already exist by checking, *sudo ip netns exec PC1 ip neigh show 192.168.10.11*, Delete the dynamic ARP entry first, *sudo ip netns exec PC1 ip neigh del 192.168.10.11 dev veth-PC1-pc*, and then repeat Step 5. Then , *execute sudo ip netns exec PC1 ip neigh show* to verify, 192.168.10.11 dev veth-PC1-pc lladdr 0a:bd:60:a8:29:e7 PERMANENT. Now PC1 will never send ARP requests for 192.168.10.11, because it already has a permanent mapping.

Part 4: Mini-Shells in Network Namespaces

In this part, the focus will be on how mini-shells can be used to explore networks interactively. This is a very powerful hands-on concept that explores networking without needing real routers or switches.

Step 1: Open mini-shells for interactive ARP testing

Open each PC and GATEWAY namespace in its own terminal

```
gnome-terminal -- sudo ip netns exec PC1 bash -c 'export PS1="PC1-mini-shell> "; exec bash'
gnome-terminal -- sudo ip netns exec PC2 bash -c 'export PS1="PC2-mini-shell> "; exec bash'
gnome-terminal -- sudo ip netns exec PC3 bash -c 'export PS1="PC3-mini-shell> "; exec bash'
gnome-terminal -- sudo ip netns exec PC4 bash -c 'export PS1="PC4-mini-shell> "; exec bash'
gnome-terminal -- sudo ip netns exec PC5 bash -c 'export PS1="PC5-mini-shell> "; exec bash'
gnome-terminal -- sudo ip netns exec PC6 bash -c 'export PS1="PC6-mini-shell> "; exec bash'
gnome-terminal -- sudo ip netns exec GATEWAY bash -c 'export PS1="GATEWAY-mini-shell> "; exec bash'
gnome-terminal -- sudo ip netns exec SW1 bash -c 'export PS1=" SW1-mini-shell> "; exec bash'
gnome-terminal -- sudo ip netns exec SW2 bash -c 'export PS1=" SW2-mini-shell> "; exec bash'
```

Step 2: Start tcpdump to monitor ARP

PCs — veth interfaces

```
sudo ip netns exec PC1 tcpdump -nni veth-PC1-pc arp &
sudo ip netns exec PC3 tcpdump -nni veth-PC3-pc arp &
sudo ip netns exec PC5 tcpdump -nni veth-PC5-pc arp &
```

```
sudo ip netns exec PC2 tcpdump -nni veth-PC2-pc arp &
sudo ip netns exec PC4 tcpdump -nni veth-PC4-pc arp &
sudo ip netns exec PC6 tcpdump -nni veth-PC6-pc arp &
```

Switches — bridge + veth interfaces

```
sudo ip netns exec SW1 tcpdump -nni br-lan1 arp &
sudo ip netns exec SW1 tcpdump -nni veth-PC1-sw arp &
sudo ip netns exec SW1 tcpdump -nni veth-PC3-sw arp &
sudo ip netns exec SW1 tcpdump -nni veth-PC5-sw arp &
sudo ip netns exec SW1 tcpdump -nni gw-lan1-sw arp &
```

```
sudo ip netns exec SW2 tcpdump -nni br-lan2 arp &
sudo ip netns exec SW2 tcpdump -nni veth-PC2-sw arp &
sudo ip netns exec SW2 tcpdump -nni veth-PC4-sw arp &
sudo ip netns exec SW2 tcpdump -nni veth-PC6-sw arp &
sudo ip netns exec SW2 tcpdump -nni gw-lan2-sw arp &
```

Step 3: Test ARP dynamically

```
# Check ARP tables before sending traffic
ip neigh
# Delete any ARP entry
sudo ip neigh del <IP> dev <interface>
# On switches (optional, see MAC forwarding table)
sudo ip netns exec SW1 bridge fdb show
sudo ip netns exec SW2 bridge fdb show
```

Step 4 Generate ARP broadcasts

```
# Within same LAN / switch, from PC1 (LAN1 → SW1)
ping -c 2 192.168.10.11 # PC3
# From PC5 (LAN1 → SW1)
ping -c 2 192.168.10.10 # PC1

# From PC1 (LAN1 → PC2, LAN2)
ping -c 2 192.168.20.10 # PC2
```

Step 5 Generate ARP broadcasts

```
# Within same LAN / switch, from PC1 (LAN1 → SW1)
```

- PC1 → entries for PC3 and PC5
- PC3 → entry for PC1
- PC5 → entry for PC1
- GATEWAY → entries for connected LAN interfaces
- PC2/PC4/PC6 → no ARP entries for PC1 unless pinged directly

Deliverable (To be uploaded on the Platform)

In this lab, we focused on ARP behavior in Layer 2 switched LANs with PCs, SW1/SW2 bridges, and a GATEWAY. ARP broadcasts allow devices to discover MAC addresses for IPs within the same LAN. The document (word/pdf, 10 pages Max) must include the following:

- Objective and Setup
 - Observe ARP behavior and broadcast traffic in Layer 2 switched LANs.
 - Namespaces: PC1–PC6, GATEWAY, SW1, SW2
 - Bridges: br-lan1 (SW1), br-lan2 (SW2)
 - IP addresses and default routes assigned
- Observations
 - ARP tables before and after ping
 - TCPDUMP outputs showing ARP requests and replies

- MAC address learning on switches

Conclusion

- Analysis / Key Points

A single compressed file named in the format “lastname.firstname-Lab1.zip”, must be uploaded on the platform (see deadlines) and must include:

- The main report document (Word or PDF)
- All developed Bash scripts